

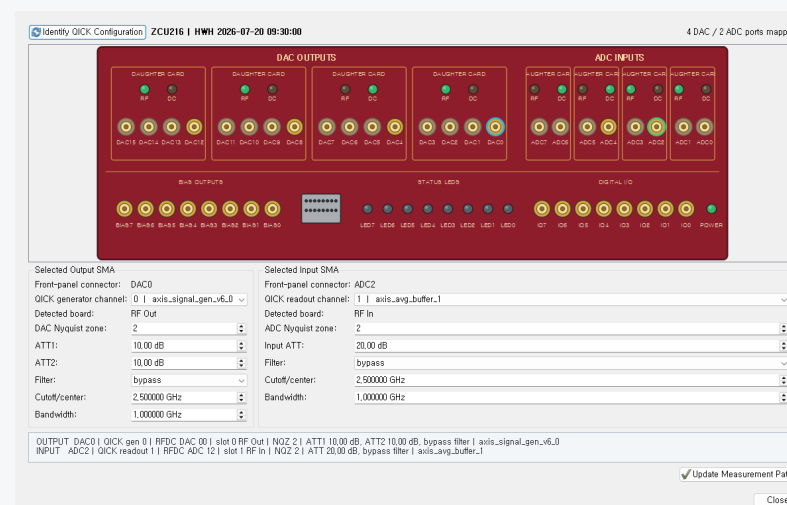
PulseGenerator GUI

User Guide for QICK-Based Pulse and RF Measurements

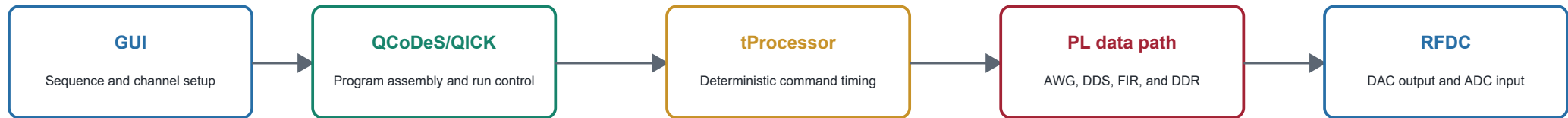
AWG tuning, multidimensional sweeps, RF output/readout, FIR-DDR capture, noise analysis, stability diagrams, S-parameters, calibration, and QCoDeS storage

Audience: experimentalists familiar with RF measurements who are new to QICK

Version 1.1 | 27 July 2026



1. What This Application Controls



Control travels left to right; captured FIR I/Q and metadata return to the host.

QICK in one paragraph

QICK combines a software-defined experiment description with FPGA data paths. The tProcessor dispatches timestamped commands. Generator IP drives RFDC DAC channels, readout IP processes RFDC ADC channels, and custom FIR/DDR logic stores long traces. The GUI builds this configuration and keeps channel routing, RF-board settings, pulse timing, and database metadata together.

- The GUI does not stream every waveform point from the PC in real time.
- Pulse commands and sweep parameters are compiled before execution.
- FIR I/Q data returns after FPGA processing and DDR capture.

Major hardware terms

- **RFDC**: integrated DAC and ADC tiles on the ZCU216.
- **tProcessor**: deterministic timing engine that issues generator and trigger commands.
- **Fabric clock**: PL clock used by AXIS generators and custom IP.
- **HWH**: hardware metadata describing instantiated IP and signal connectivity.
- **FIR-DDR**: filtered, decimated capture path that writes I/Q samples into PL DDR.

Firmware assumption

This guide covers the qstl_awg_tuning_fir 1 MSPS image and the HWH-identified 50 kSPS variant. A different BIT/HWH pair can expose a different channel map, FIR rate, trigger-delay unit, or feature set.

2. Installation and First Connection

Recommended host environment

Use the repository requirements in a dedicated Conda environment. The GUI needs PyQt5, pyqtgraph, NumPy, QCoDeS, Pyro, and the matching local QICK package.

Typical launch

```
conda activate qick_test
python DCWaveformGeneratorGUI/DCWaveform_Generator.py
```

QICK server

Start the Pyro nameserver and QICK server on the ZCU216, then enter the host, nameserver port, and proxy name in AWG Tuning > Experiment.

Before energizing a device

- 1. Load firmware** - Confirm the intended BIT/HWH pair is active.
- 2. Identify configuration** - Open the Front Panel and read live HWH routing.
- 3. Verify daughter cards** - Confirm RF_Out, DC_Out, RF_In, or DC_In for each connector.
- 4. Set RF protection** - Start with high attenuation and a conservative DAC gain.
- 5. Dry run** - Use Show Program before the first hardware execution.

Clock defaults

The GUI defaults the AWG fabric clock and tProcessor time base to 300 MHz. Keep these values consistent with the deployed firmware and measured timing.

tProcessor time



Instruction timestamp

AWG fabric



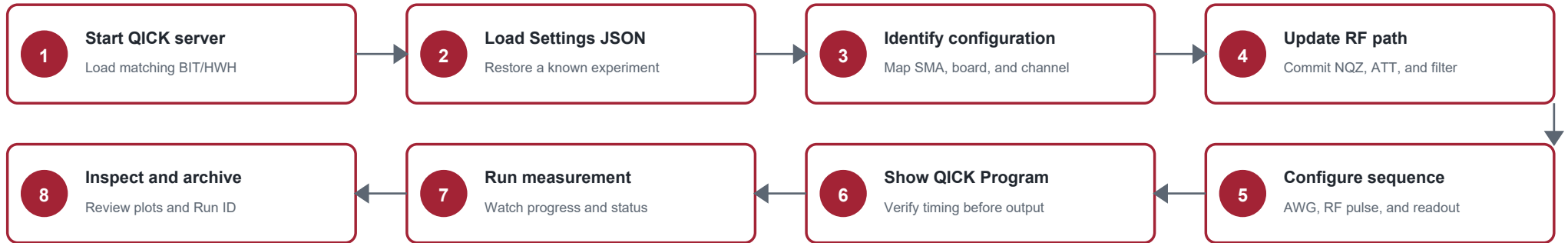
One AXIS word con

FIR DDR output



HWH-selected store

3. Quick Start: Setup Order



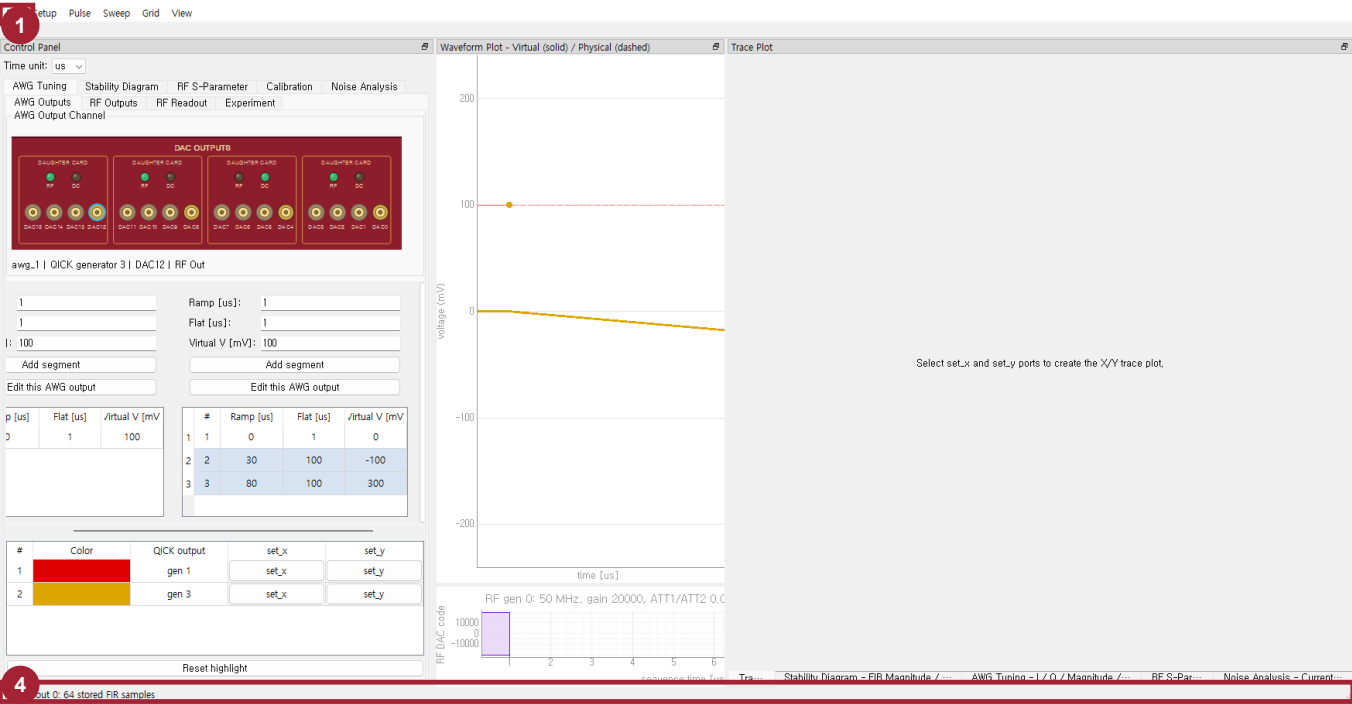
- 1 Start the hardware service**
Start the Pyro nameserver and QICK server with the intended BIT/HWH pair.
- 2 Load a known JSON**
Use File > Load Settings JSON to restore waveform, RF, database, and display settings.
- 3 Identify QICK Configuration**
Read live HWH routing and detected daughter-card types.
- 4 Update Measurement Path**
Commit SMA mapping, channel, board, NQZ, attenuation, filter, and losses.

- 5 Configure the sequence**
Build AWG segments, RF output pulses, and the FIR-DDR readout window.
- 6 Show QICK Program**
Inspect generated instructions and timing before energizing the DUT.
- 7 Run**
Start the requested experiment and watch progress/status until completion.
- 8 Inspect and save**
Review plots, QCoDeS Run ID, metadata, and the status line before changing hardware.

Recommended practice

Complete steps 1-6 with conservative gain and high attenuation. Save a new JSON only after the channel map and timing have been verified.

4. Main Window, Save, and Load



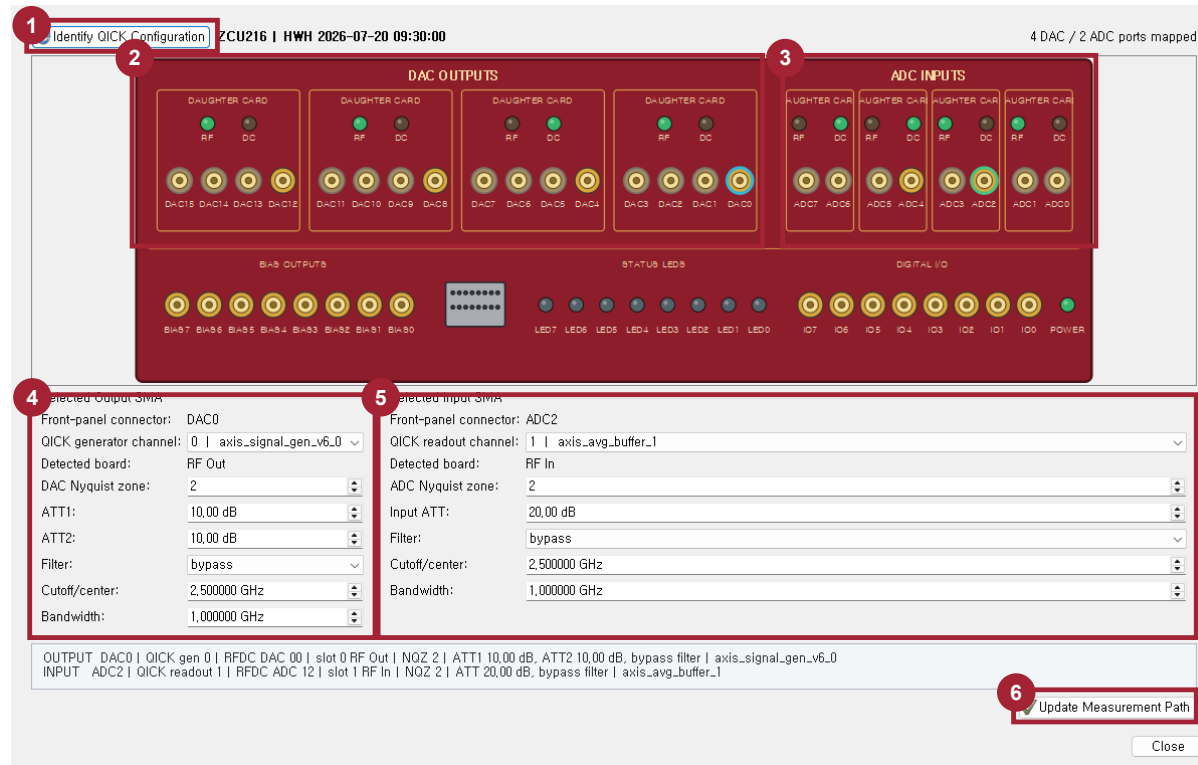
- 1 **Open File**
Click File in the top-left menu bar.
- 2 **Save**
Click Save Settings JSON... or press Ctrl+S.
- 3 **Load**
Click Load Settings JSON... or press Ctrl+O.
- 4 **Confirm**
Read the bottom status bar for the saved or loaded path.

JSON contents

A settings file stores AWG vertices, sweeps, cross-capacitance, RF output/readout, shared RF path, calibration and database paths, clocks, compensation, and display preferences. Loading an older schema fills new fields with safe defaults; saving again upgrades it.

5. HWH-Backed Front Panel

Follow the numbered controls from live HWH discovery to a committed RF path.



1 Identify

Click Identify QICK Configuration to read HWH and daughter-card IDs.

2 Select output SMA

Click the physical DAC connector that drives the DUT.

3 Select input SMA

Click the physical ADC connector returning from the DUT.

4 Configure output

Confirm generator, board, NQZ, ATT1/ATT2, and filter.

5 Configure input

Confirm readout, board, NQZ, input ATT or DC gain, and filter.

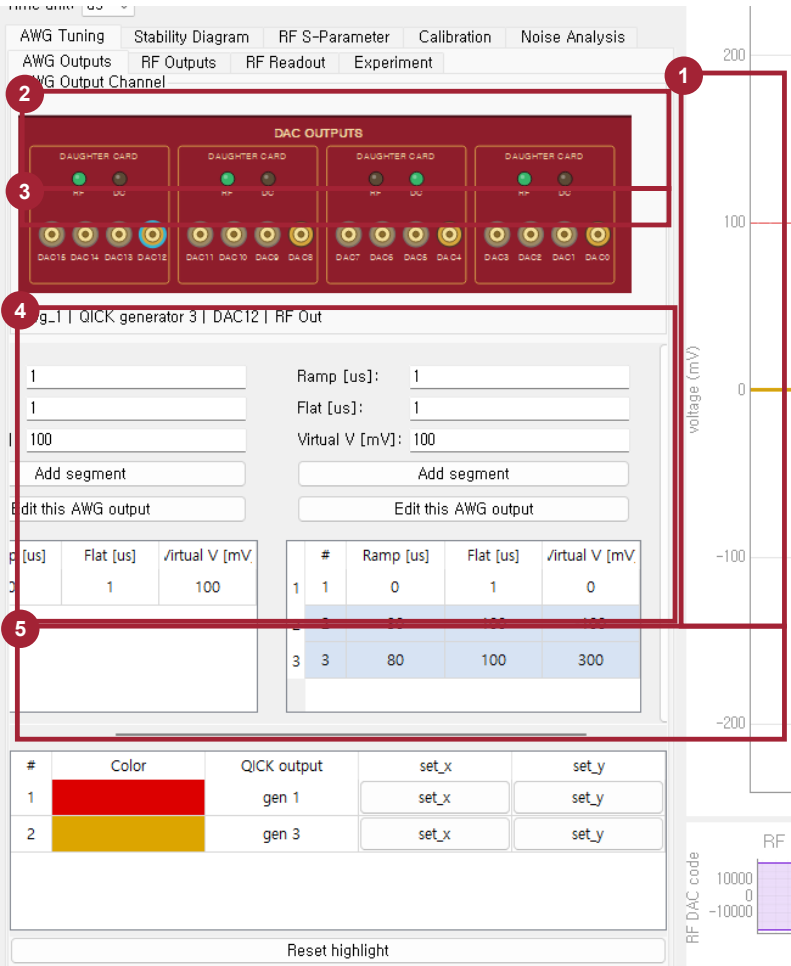
6 Commit path

Click Update Measurement Path to propagate settings to all measurement tabs.

Nyquist zones

NQZ changes RFDC reconstruction or sampling-zone configuration, not the logical sweep frequency. Confirm the analog band and anti-alias filtering before selecting zone 2.

6. AWG Waveform Authoring



- 1 Add Port**
Create up to eight AWG output editors.
- 2 Enter segment values**
Set ramp time, flat time, and virtual target voltage.
- 3 Add segment**
Append the entered SET/RAMP pair to the selected port.
- 4 Edit a row**
Right-click a segment row to insert, delete, or configure a voltage sweep.
- 5 Choose plot axes**
Use set_x and set_y to select traces for the X/Y plot.

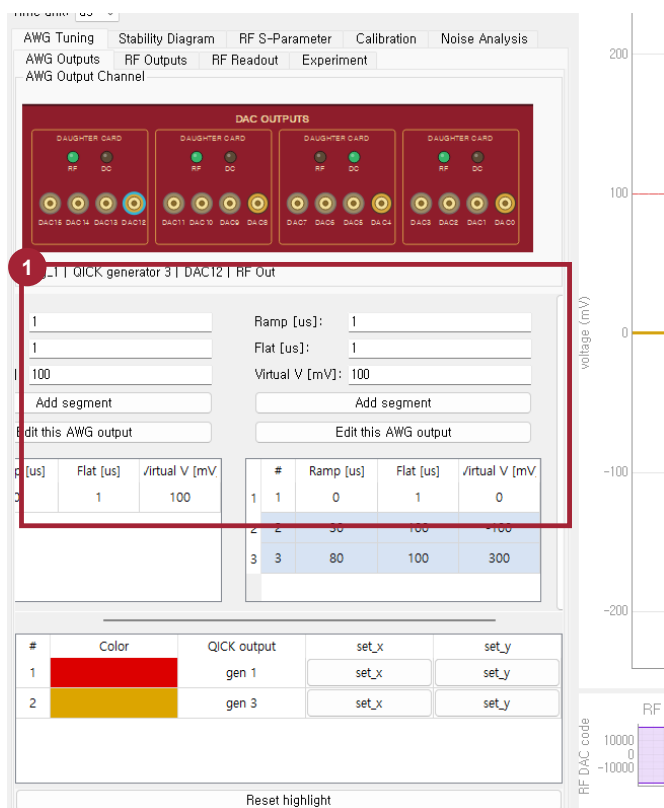
Cross-capacitance

Use Pulse > Virtual-to-physical cross-capacitance matrix... to map virtual electrodes to physical DAC voltages. Review solid virtual and dashed physical traces before execution.

Grid

Use Grid > Configure grid... and optionally enable Snap dragged points.

7. Voltage Sweeps and Bias-T Compensation



- 1 **Right-click a SET row**
Use the boxed segment table, not the waveform line.
- 2 **Open Voltage sweep**
Choose Voltage sweep > Configure sweep... from the context menu.
- 3 **Enter the range**
Set start voltage, stop voltage, and point count, then accept the dialog.
- 4 **Verify the envelope**
The GUI redraws the extreme traces; adjacent ramps follow the swept SET automatically.

Multiple voltage sweeps

Configure additional output/SET rows the same way. The GUI creates a Cartesian product and nests repetitions inside each sweep point.

DC compensation

In AWG Tuning > Experiment, enable Bias-T compensation, choose DC compensation, then select fixed voltage or fixed time. The complementary value is calculated before program generation.

Filter compensation

Choose Filter compensation and enter tau. Flat segments receive target/tau slew-rate correction. Use Show QICK Program before running.

Safety

Compensation can increase peak voltage or sequence duration. Verify the physical traces and full-scale limits before output.

8. Independent RAMP-Rate Sweeps

Each selected RAMP segment becomes its own duration axis; the target SET voltage remains unchanged.



awg_1 | QICK generator 3 | DAC12 | RF Out

p [us]	Flat [us]	/virtual V [mV]
1	1	100

Add segment

Edit this AWG output

#	Ramp [us]	Flat [us]	/virtual V [mV]
1	0	1	0
2	30	100	-100
3	80	100	300

RAMP segment: ramp_1_to_2
Current duration: 80 us
Nominal voltage change: -400 mV
Start duration: 80,00000000 us
Stop duration: 80,00000000 us
Sweep point count: 4
Nominal RAMP rates: -10 to -5 mV/us (nominal endpoint levels)
Cartesian total: 3 x 4 = 12 points

OK Cancel

- 1 Create distinct RAMPs**
Add at least three SET rows so ramp_0_to_1 and ramp_1_to_2 both exist.
- 2 Open the first axis**
Right-click the destination SET row, then choose RAMP rate sweep > Configure sweep...
- 3 Enter durations**
Set start duration, stop duration, and point count. The dialog reports the derived nominal mV/us range.
- 4 Repeat on another RAMP**
Right-click a different destination SET row and configure its independent duration axis.
- 5 Check the product**
Cartesian total multiplies every active RAMP, voltage, and RF-duration axis. Repetitions run inside each point.
- 6 Select map axes**
In AWG Tuning > Experiment, choose the desired RAMP duration axes for X and Y. Units are microseconds.

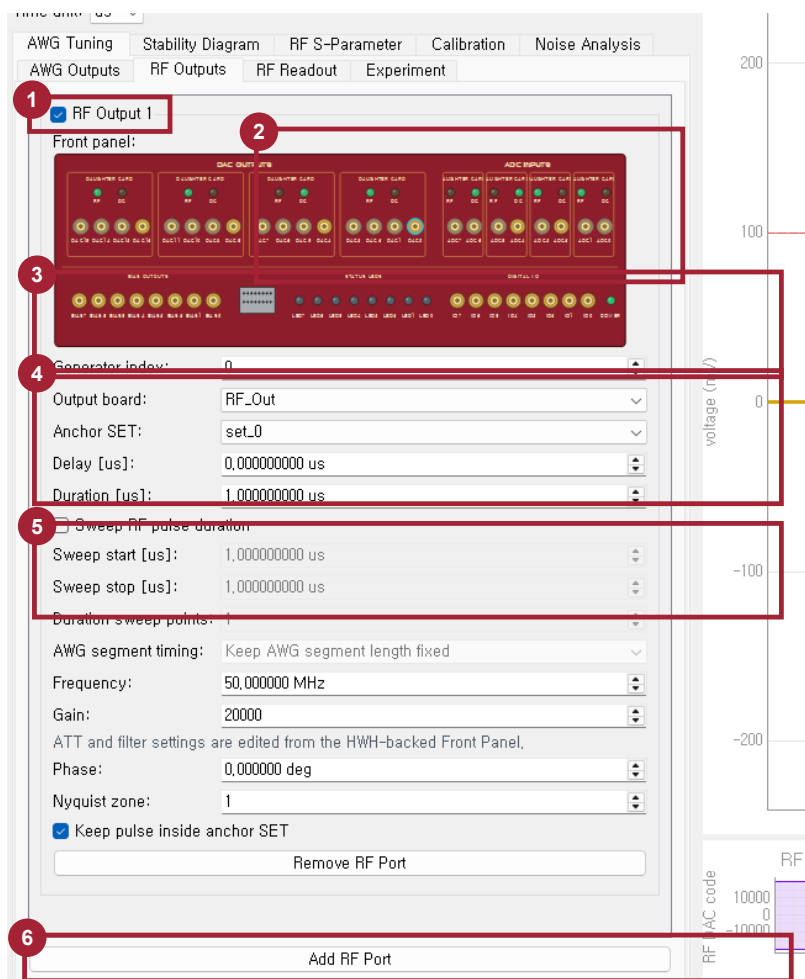
Execution model

The tProcessor uses one compact DMEM coefficient table per RAMP segment. Memory scales with the sum of axis sizes, not the full Cartesian point count.

Bias-T compatibility

Filter compensation supports multiple RAMP-rate axes. DC Bias-T compensation currently permits one RAMP-rate axis and rejects a multi-axis request explicitly.

9. RF Output Channels

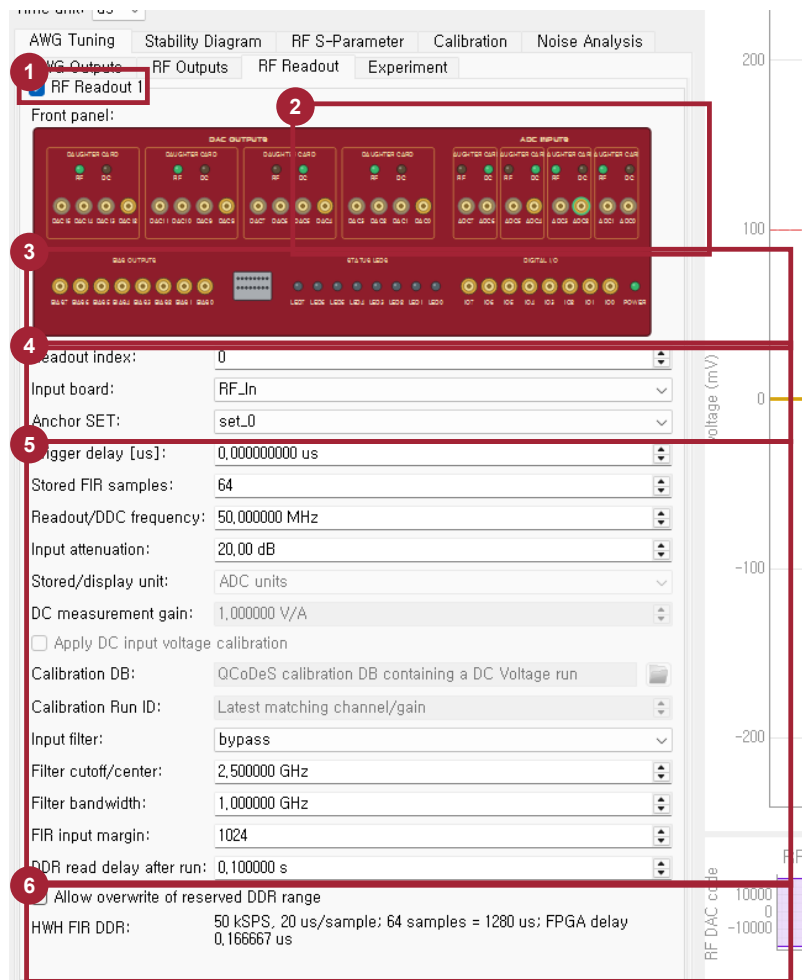


- 1 **Enable RF Output**
Check the channel group before editing or running.
- 2 **Open the Front Panel**
Click the panel preview to select SMA, board, ATT, filter, and NQZ.
- 3 **Select routing**
Set generator index, output board, and anchor SET.
- 4 **Set the pulse**
Enter delay, duration, frequency, and gain.
- 5 **Finish timing**
Set phase/NQZ and keep the pulse inside its anchor SET when required.
- 6 **Add another RF port**
Use Add RF Port for another generator channel.

Gain limit

Start with conservative gain and high attenuation. The software rejects settings above the configured maximum.

10. RF Readout and FIR-DDR Capture



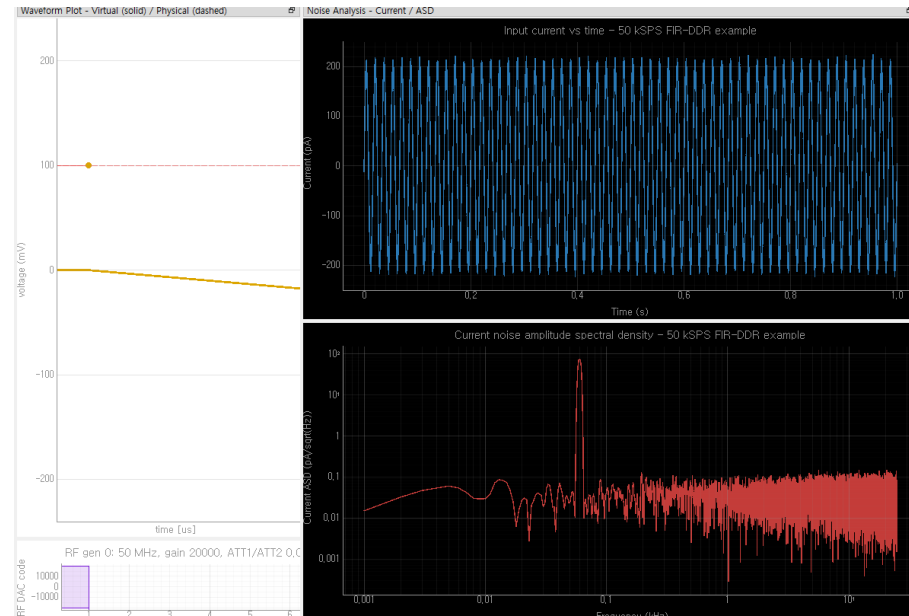
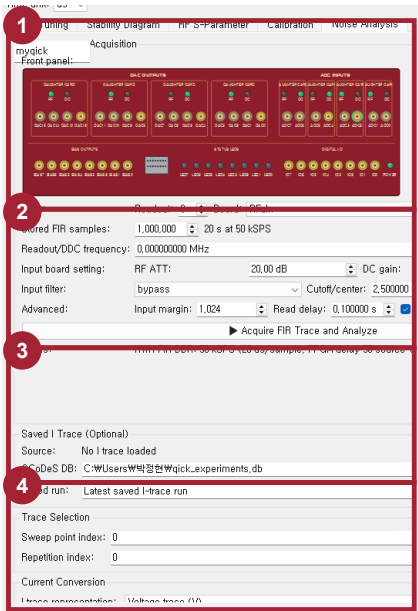
- 1 **Enable RF Readout**
Check the readout group.
- 2 **Open the Front Panel**
Select ADC SMA, board, NQZ, input ATT or DC gain, and filter.
- 3 **Choose routing**
Set readout index, input board, and anchor SET.
- 4 **Define capture**
Set trigger delay, stored sample count, and DDC frequency. The HWH label shows 1 MSPS or 50 kSPS.
- 5 **Set analog conversion**
For DC_In, optionally convert FIR I/Q to current using the configured gain.
- 6 **Set FIR-DDR safety**
Choose FIR warm-up margin and enable overwrite only for a safe reserved range.

Stored data

The deployed design filters continuously and stores post-FIR I/Q at the HWH-selected rate. The 1 MSPS image uses the legacy software warm-up model. The 50 kSPS image reports an FPGA trigger-delay register and software does not add the FIR group delay to tProcessor timing.

11. Noise Analysis

Acquire an independent FIR-DDR I trace or reload a saved QCoDeS run, then inspect current and amplitude spectral density.



1 Select an independent input

Use the tab-local QICK connection and front panel to choose ADC SMA, board, NQZ, ATT or DC gain, filter, and DDC frequency.

2 Set FIR acquisition

Enter stored sample count, margin, and DDR read delay. The duration and sampling rate come from HWH.

3 Acquire or reload

Click Acquire FIR Trace and Analyze, or select a QCoDeS DB and saved Run ID.

4 Choose conversion

Display raw ADC units, calibrated voltage, or current using transimpedance gain and optional DC-input calibration.

5 Configure the spectrum

Choose window, detrend, and optional NFFT. The ASD is one-sided and uses the actual stored sample rate.

6 Interpret the plots

The upper plot is current versus time; the lower log-log plot is current ASD in $A/\sqrt{Hz}$.

Large traces

Numerical analysis uses the complete trace. The plot display decimates very large arrays for responsiveness, so a 10,000,000-sample capture can be analyzed without drawing every point.

12. Running an AWG Tuning Experiment

AWG Tuning | Stability Diagram | RF S-Parameter | Calibration | Noise Analysis

AWG Outputs | RF Outputs | RF Readout | Experiment

1 QCoDeS DB file: C:\Users\박정현\qick_experiments.db

2 Experiment name: QICK pulse experiment

Sample name: PulseGenerator

AWG full scale (+/-): 2500.000000 mV

Repetitions per sweep point: 1

AWG 2D sweep map

X sweep variable: ramp_0_to_1 RAMP duration (rate derived) | 20 to 40 us | 3 po

Y sweep variable: ramp_1_to_2 RAMP duration (rate derived) | 40 to 80 us | 4 po

12 Cartesian points; I/Q are averaged over repetitions and FIR samples.

☐ Bias-T compensation

Compensation type: DC compensation

DC control mode: Fixed voltage (adjust time)

DC voltage: 250.000000 mV

DC time: 1.000000 us

Filter time constant (tau): 100.000000 us

Notes: Optional experiment notes

5 Run QICK Experiment

1 Show QICK Program

Status: Ready

voltage (mV)

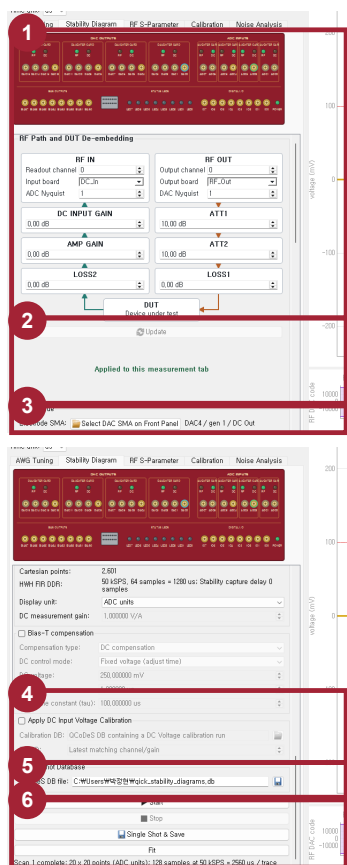
RF DAC code

- 1 Connect**
Enter QICK host, nameserver port, and proxy name.
- 2 Choose storage**
Select the QCoDeS DB, experiment name, and sample name.
- 3 Confirm timing**
Set fabric/tProcessor clocks, full scale, AWG indices, and repetitions.
- 4 Show QICK Program**
Inspect assembly and timing before hardware execution.
- 5 Run QICK Experiment**
Start the full sweep/repetition job and watch progress/status.

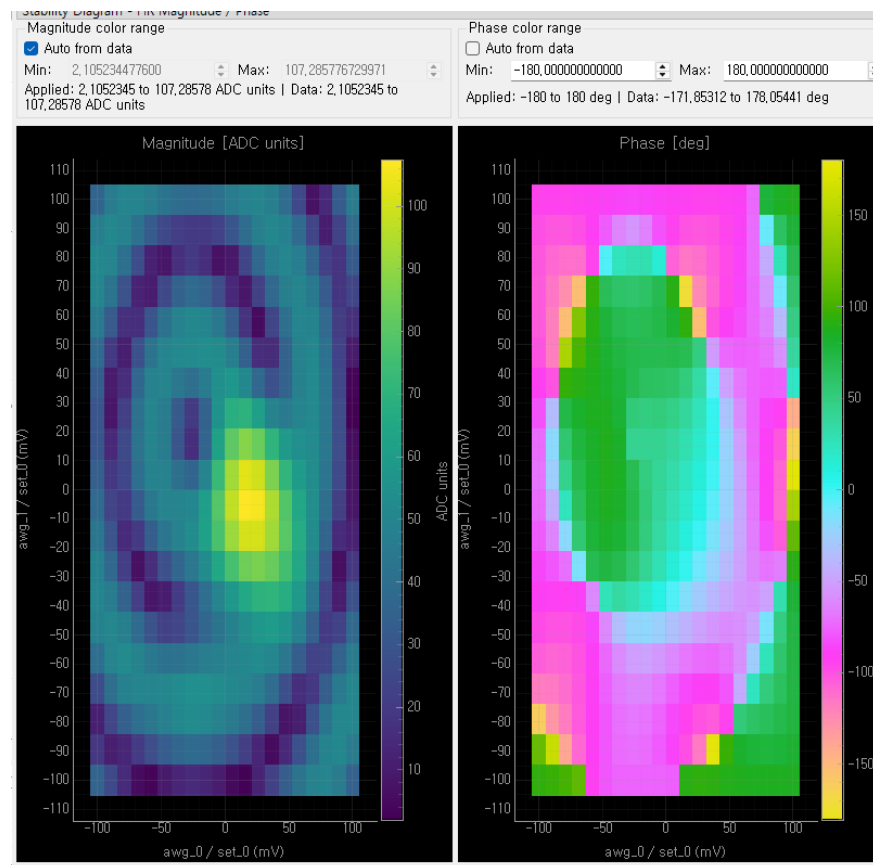
Large data sets

Write to a local SSD, checkpoint SQLite WAL at completion, then copy to synchronized storage. Traces are stored as arrays rather than one SQL row per sample.

13. Stability Diagram



- 1 **Set measurement path**
Choose the output/readout channels and commit their settings for this tab.
- 2 **Set X electrode**
Select the physical DAC SMA, then enter start/stop voltage and point count.
- 3 **Set Y electrode**
Select the second physical DAC SMA and its Cartesian sweep range.



- 4 **Set acquisition**
Choose repetitions, FIR trace length, modulation frequency/gain, display units, and optional DC conversion.
- 5 **Choose single-shot DB**
This database is used only by Single Shot & Save.
- 6 **Start or save**
Start repeats without DB writes; Stop ends after the active scan; Single Shot & Save stores one map.

14. RF S-Parameter Sweep



1 Set RF path

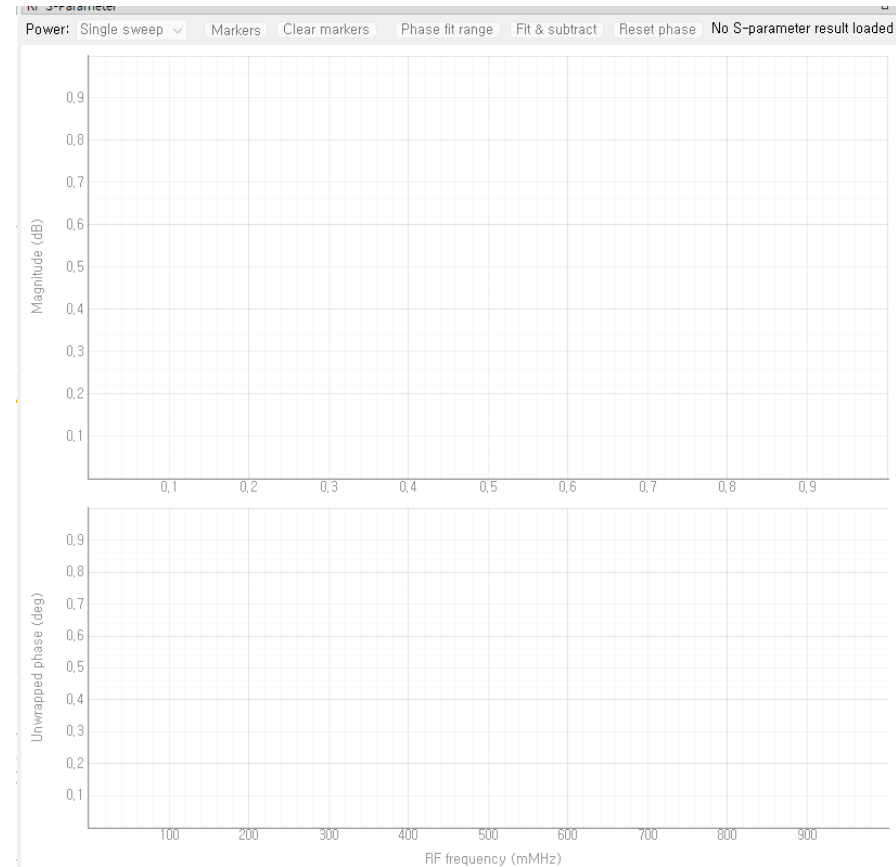
Commit output/readout SMA, boards, NQZ, ATT, filters, and de-embedding terms.

2 Set frequency sweep

Enter start/end frequency, point count, gain or target power, and scan time.

3 Optional correction

Enable frequency-response compensation and/or the outer software power sweep.



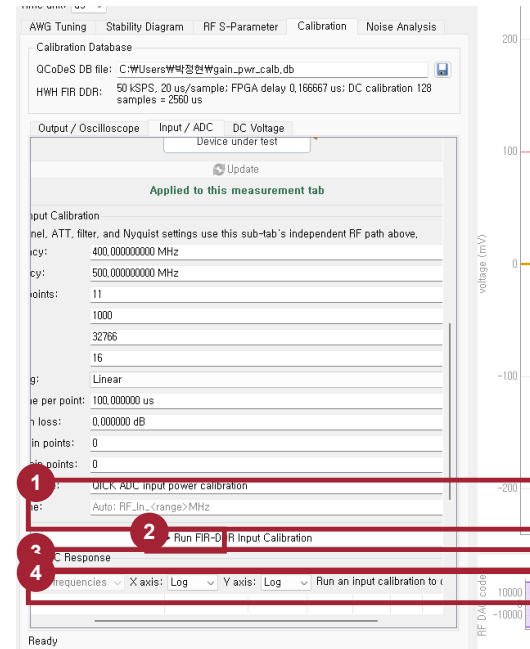
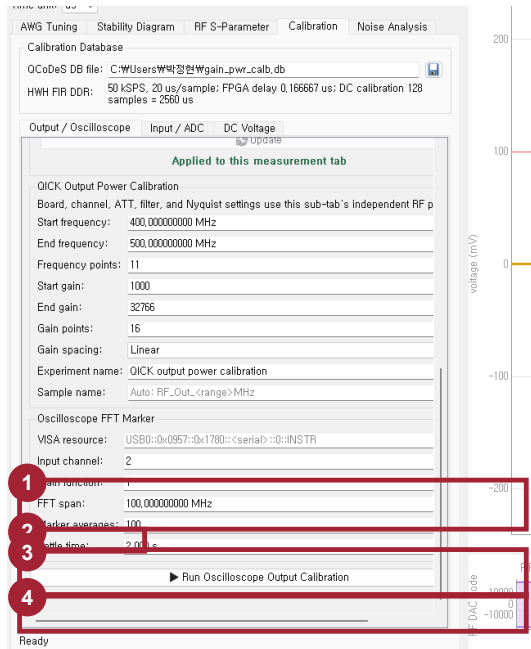
4 Configure capture and DB

Set FIR margin, DDR address/stride/overwrite, and the independent S-parameter DB.

5 Run or reload

Click Run RF S-Parameter Sweep, or select a Run ID and click Load Saved Run.

16. Calibration Workflows



- 1 **Choose calibration DB**
Select the QCoDeS gain/power database.
- 2 **Output / Oscilloscope**
Open this tab for absolute output-power calibration.
- 3 **Set scope controls**
Enter frequency/gain axes, VISA resource, FFT marker, averaging, and settle time.
- 4 **Run output calibration**
Click Run Oscilloscope Output Calibration.

Physical S21

Compatible output and input calibrations report calibrated DUT power. Otherwise the result is relative ADC response, not absolute VNA-grade S21.

- 1 **Use the same DB**
Input and output calibration runs share the selected file.
- 2 **Input / ADC**
Open this tab for input-power versus FIR-DDR ADC response.
- 3 **Set scan metadata**
Confirm frequency/gain scan and experiment/sample names.
- 4 **Run input calibration**
Click Run FIR-DDR Input Calibration; use the plot controls afterward.

DC voltage calibration

The DC Voltage tab sweeps DC_Out at 0 MHz into DC_In and fits $ADC-I = \text{offset} + \text{slope} \times \text{voltage}$. Applying the Run ID converts later DC and Noise Analysis data to volts.

17. QCoDeS Data Layout

Per-shot coordinates

- Voltage-sweep coordinates identify the AWG output and SET segment, not generic sweep0/sweep1 labels.
- Each RAMP-rate axis uses its own segment-specific coordinate and stores ramp_duration_us.
- RF pulse-duration axes are also named for their RF output and anchor segment.
- repetition_index identifies the repeated shot inside one Cartesian sweep point.
- I trace and Q trace are stored as full-length one-dimensional arrays.

AWG vertices

- Each AWG channel stores virtual and physical vertex traces separately.
- Vertex time is reconstructed from paired time and voltage arrays; it is not exposed as a sweep selector.

Metadata

- BIT/HWH identity, clocks, channel map, board types, NQZ, attenuation, filters, and RF settings.
- Pulse sequence, cross-capacitance matrix, compensation mode, sweep definition, and repetitions.
- FIR profile/rate, sample count, trigger-delay units, scan timing, calibration Run IDs, and de-embedding terms.

Plottr-inspectr

Select I trace or Q trace as the dependent array and sample index as its array axis. Use sweep coordinates and repetition_index to choose or compare shots.

Noise Analysis

The tab can acquire a fresh FIR-DDR trace or reload a saved QCoDeS I-trace run. Spectrum processing does not modify the stored trace.

Database size

Array-per-trace storage avoids millions of scalar SQL rows. Total size still scales with Cartesian points x repetitions x samples x two channels, so estimate storage before a large run.

18. Troubleshooting and Operating Checklist

Common problems

- **Channel appears wrong:** re-run Identify QUICK Configuration and confirm HWH matches BIT.
- **First trace is noisy:** verify arm-before-trigger order, readout warm-up, and trigger offset. Do not assume DDR is the only source.
- **Timing scales differ:** verify the Setup clocks and the HWH-selected FIR profile against deployed firmware.
- **50 kSPS trigger-delay error:** use the delay units reported by HWH; do not add the FIR group delay to tProcessor scheduling.
- **Attenuator does not respond:** confirm daughter-card seating and detected board type before changing software.
- **No Numpy._core through Pyro:** return plain lists/scalars across RPC or align NumPy serialization versions.

Pre-run checklist

1. **Firmware** - Correct BIT and HWH loaded.
2. **Routing** - Physical SMA, logical channel, board type, and NQZ confirmed.
3. **Protection** - ATT/filter/gain and voltage full-scale reviewed.
4. **Timing** - Pulse, readout, FIR margin, and compensation windows checked.
5. **Storage** - DB path, free space, sweep size, and repetition count checked.
6. **Dry run** - Generated program inspected before energizing the DUT.

Support information

When reporting a problem, include the saved JSON, software commit, BIT/HWH pair, HWH channel summary, QCoDeS Run ID, complete traceback, and the exact settings that produced the issue.